

10(a)	any four from:		4																														
	<table><tr><td></td><td>structure</td><td>function in photosynthesis</td><td></td></tr><tr><td>1</td><td>stroma</td><td>site of, (enzymes in) light-independent stage / Calvin cycle</td><td>;</td></tr><tr><td>2</td><td rowspan="2">thylakoid membranes</td><td>site of, light-dependent stage / photosystems / (named) pigments / oxygen-evolving complex</td><td>;</td></tr><tr><td>3</td><td>contains, electron transport chain / ETC / ATP synthase / electron carriers</td><td>;</td></tr><tr><td>4</td><td><u>grana / granum</u></td><td>to contain many, (named) pigment / photosystems or increase surface area for light (energy) absorption</td><td>;</td></tr><tr><td>5</td><td>thylakoid, space / lumen</td><td><i>ref. to</i> setting up proton gradient / contains a high concentration of protons</td><td>;</td></tr><tr><td>6</td><td>DNA</td><td>codes for (named) proteins / ribosomes</td><td>;</td></tr><tr><td>7</td><td>ribosomes</td><td>translation (of mRNA) into / production of, (named) proteins</td><td>;</td></tr></table>		structure	function in photosynthesis		1	stroma	site of, (enzymes in) light-independent stage / Calvin cycle	;	2	thylakoid membranes	site of, light-dependent stage / photosystems / (named) pigments / oxygen-evolving complex	;	3	contains, electron transport chain / ETC / ATP synthase / electron carriers	;	4	<u>grana / granum</u>	to contain many , (named) pigment / photosystems or increase surface area for light (energy) absorption	;	5	thylakoid, space / lumen	<i>ref. to</i> setting up proton gradient / contains a high concentration of protons	;	6	DNA	codes for (named) proteins / ribosomes	;	7	ribosomes	translation (of mRNA) into / production of, (named) proteins	;	
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10(b)(i)	B = 0.42 ;		1																														
10(b)(ii)	A = carotene B = chlorophyll b ;		1																														

Question	Answer	Marks
10(b)(iii)	any two from: 1 act as accessory pigments ; Ignore secondary pigments 2 absorbs light wavelengths not absorbed by, reaction centre / primary pigment / chlorophyll a ; 3 so extends the range of wavelengths absorbed ; 4 pass energy to, reaction centre / primary pigment / chlorophyll a ;	2